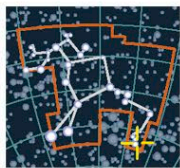


EMISSION NEBULA

IC 2944



CATALOG NUMBER
IC 2944

DISTANCE FROM SUN
5,900 light-years

MAGNITUDE 4.5

CENTAURUS

Between the constellations Crux and Centaurus lies the bright, busy star-forming nebula IC 2944. This nebula is made up of dust and gas that is illuminated by a loose cluster of massive young stars. IC 2944 is perhaps best known for the many Bok globules that are viewed in silhouette against its backdrop. Bok globules are thought to be cool, opaque regions of molecular material that will eventually collapse to form stars. However, studies of the globules in IC 2944 have

THACKERAY'S GLOBULES

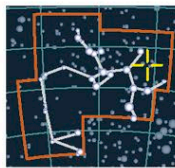
The Bok globules in IC 2944 were first observed in 1950 by South African astronomer A. D. Thackeray. This globule has recently been shown to be two overlapping clouds.



revealed that the material of which they are composed is in constant motion. This may be caused by radiation from the loose cluster of massive young stars embedded in IC 2944. The stars' ultraviolet radiation is gradually eroding the globules, and it is possible that this could prevent them from collapsing to form stars. In addition to radiation, the stars also emit strong stellar winds that send out material at high velocities, causing heating and erosion of interstellar material. The largest Bok globule in IC 2944 (below) is about 1.4 light-years across, with a mass about 15 times that of the Sun.

EMISSION NEBULA

Sharpless 29



CATALOG NUMBER
SH 2-29

DISTANCE FROM SUN
4,100 light-years

MAGNITUDE 8

SAGITTARIUS

Lying close to the brighter and more famous Lagoon nebula in Sagittarius, Sharpless 29 is a compact region of star formation where the turbulence caused by radiation from newborn stars can be seen in detail. Stars within the nebula are around 2 million years old, and their fierce ultraviolet light excites nearby hydrogen gas to create a series of glowing halos. Meanwhile, radiation and stellar winds combine to blow bubbles within the nebula and ripple its silhouetted dust lanes.

BINARY BUBBLE

Near the center of this image, a newborn binary system has blown a large cavity in the nebula, creating a bright shock front on its upper edge.



EMISSION NEBULA

Trifid Nebula



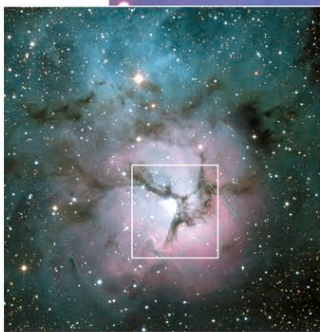
CATALOG NUMBER
M20

DISTANCE FROM SUN
7,600 light-years

MAGNITUDE 6.3

SAGITTARIUS

This emission nebula is one of the youngest yet discovered. It was first called the Trifid Nebula by English astronomer John Herschel because of its three-lobed appearance when seen through his 18th-century telescope. The nebula is a region of interstellar dust and gas being illuminated by stars forming within it. It spans a distance of around 50 light-years. The young star cluster at its center, NGC 6514, was formed only about 100,000 years ago. The Trifid's lobes, the brightest of which is actually a multiple system, are created by dark filaments lying in and around the bright nebula. The whole area is surrounded by a blue reflection nebula, particularly conspicuous in the upper part, where dust particles disperse light.



HEART OF THE TRIFID

The main image, spanning about 20 light-years, reveals details of the NGC 6514 star cluster and the filaments of dust weaving through the Trifid Nebula. A wider view (above) shows the full breadth of the nebula.

